Shravan Paladugula

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Education

University of California, San Diego — La Jolla, CA
B.S. in Computer Engineering, Jacobs School of Engineering — *Expected June 2029*
- **Coursework:** Linear Algebra (MATH 18), Object-Oriented Programming in Java (CSE 11)

Mountain House High School — Mountain House, CA
GPA: 4.64 (weighted) | Rank: 6/615 | AP Scholar with Distinction | SAT: 1460 — *Graduated May 2026*

Research Experience

Undergraduate Researcher — Sept. 2026 – Present
Mobile & Cyber-Physical Systems Lab (Prof. Xinyu Zhang), UC San Diego — *La Jolla, CA*
- Build an **agentic RF circuit-design** pipeline automating antenna, filter, and metamaterial design to **PCB**

Student Researcher, UAS Synthetic Aperture Radar — Summer 2025
MIT Beaver Works Summer Institute (1 of 24 selected nationally) — *Cambridge, MA*
- Integrated a **SAR payload** on a custom UAV; wrote **backprojection imaging** in MATLAB, Python, C++
- Tuned Pixhawk flight controllers with **PID stabilization**; team placed **1st** in the final competition

Projects

CHEHRA | *KiCad, Orange Pi, OpenCV, Supabase* — 2026
- Designed **custom PCBs** and enclosure for an edge-AI facial-recognition attendance terminal with cloud sync
- Hit **96% accuracy** over 250+ check-ins in 2 pilot classrooms; under evaluation for government deployment

PillWatch | *ESP32, Raspberry Pi, C++, Node.js* — Best Hardware Hack, CruzHacks
- Wrote **ESP32 firmware** for PID-controlled dispensing, linked via **UART** to a Pi doing face-recognition ID
- Built a Node.js backend and caregiver portal for remote scheduling, live dispensing logs, and analytics

CONJR | *Orange Pi 5 Pro, Llama 3.3 70B, Three.js, Moonraker API* — 2026
- Voice-to-3D-print kiosk: an **LLM turns speech into OpenSCAD**; Three.js previews, Moonraker prints

Shakti | *Python, OpenCV, MediaPipe* — 1st Overall, BISV Hacks 2026
- Real-time **pose-estimation** engine that scores lifting form and gives live corrective feedback

Leadership

President & Lead Director — 2023 – 2026
MHHS Hackathon Club / MountainHacks — *Mountain House, CA*
- Grew club to **500+ members**; ran a 200+ hacker event with 10+ sponsors (incl. GitHub), **$150K in prizes**

Lead Programmer & Team Founder — 4 Seasons
VEX Robotics Team 19359 & FTC Team 23649 — *Mountain House, CA*
- Wrote **PID control**, motion profiling, and sensor fusion; top-75 national Skills rank; founded team 19359B

Awards

Hackathons: 1st Overall, BISV Hacks 2026; 1st Open Innovation, AeroHacks 2026; Best Hardware Hack, CruzHacks
Competitions: TSA Engineering Design State Champion; DECA State Champion & Top 20 ICDC
Certifications: Oracle Certified Associate, Java SE 8 Programmer I

Technical Skills

Languages: C, C++, Python, Java, MATLAB, JavaScript
Embedded/Hardware: ESP32, Raspberry Pi, Pixhawk, UART, PID Control, KiCad PCB Design, RF/SAR
Software/AI: OpenCV, MediaPipe, LLM Integration, Node.js, React Native, Three.js, Supabase, Git